Correcting the Analytical Null for Discrete Mutual Information

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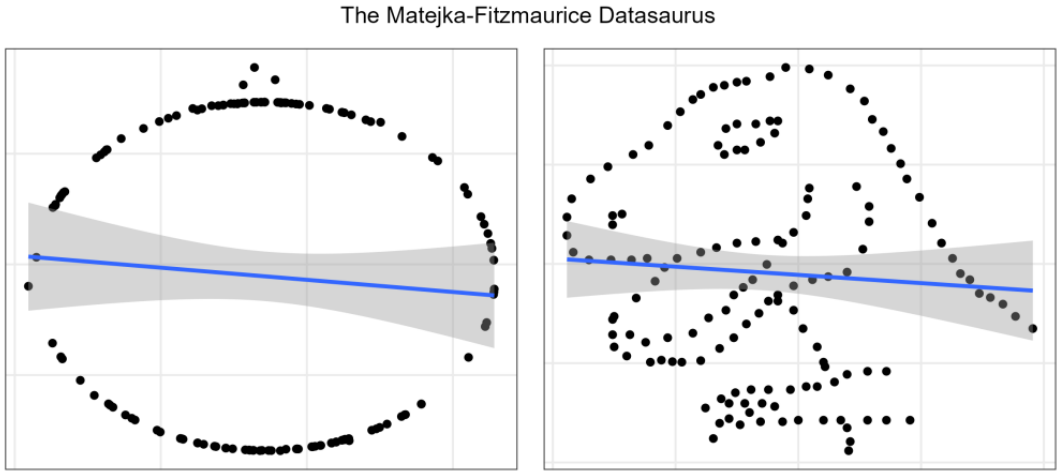
What is Mutual Information?

Mutual information is a measure from information theory that quantifies how much knowing one variable reduces uncertainty about another. A larger value indicates a stronger statistical dependence between the two.

$$I(X;Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)}$$

Fun Fact!
If X and Y are Gaussian, Mutual Information is a Function of Pearson Correlation

Measures Non-Linear Relationships[1]



Pearson Corr: -0.06
Mutual Info: 0.79

Pearson Corr: -0.07
Mutual Info: 0.56

Model-free measure

We only need to know the joint probability distribution of X and Y

↓

Estimate $p(x)$, $p(y)$, and $p(x,y)$ using data

↓

Use Plugin Estimator For Discrete Data

e.g. $\hat{p}(X = x_j) = \frac{n_j}{N}$

1. Image sourced from https://mattiheino.com/2019/05/10/correlation/. Numbers are for illustrative purposes.

Motivation: Significance Testing

Two principal approaches are commonly used to test the statistical significance of Mutual Information. Each approach involves distinct trade-offs that affect the speed and reliability of the resulting inference.

Resampled Null Distribution

How It Works

- Resample (or shuffle) one variable to destroy X-Y dependence

X	Y
a	a
a	a
b	b
b	b
c	c
c	c

→

X	Y
a	a
a	c
b	a
b	b
c	b
c	c

- Recompute MI for each resample to build empirical null
- Compare observed MI against empirical null percentiles

✓ Strengths

- Industry Standard for Significance Testing
- Easy to implement and understand

✗ Weaknesses

- Computationally expensive (thousands of resamples)
- Cost of significance testing dominates the cost of estimation

Analytical Null Distribution

How It Works

- Derive MI's null distribution mathematically

$$2N\hat{I}(X;Y) \approx N\sum_{x,y}\frac{\delta(x,y)^2}{\hat{p}(x)\hat{p}(y)} \xrightarrow{d} \chi^2_{(|X|-1)(|Y|-1)}$$

- Use asymptotic theory (e.g. χ^2) for p-values
- No resampling required, only need alphabet size and sample size

✓ Strengths

- Computationally fast: closed-form p-values
- Only need to know alphabet size and sample size

✗ Weaknesses

- Works poorly in small samples or skewed distribution
- Need dense sampling across the joint distribution of X and Y

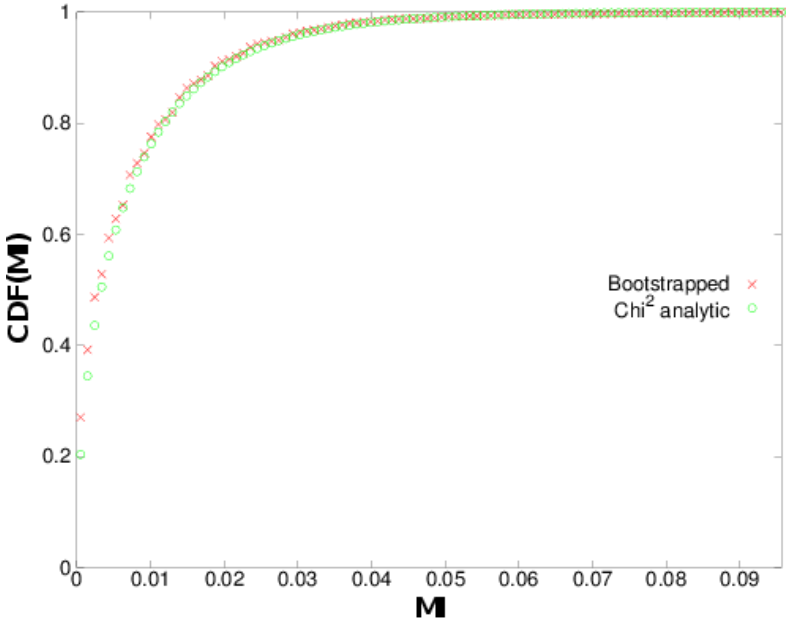
Motivation: Analytical Null Shortfall

The standard chi-square analytical null is convenient but can be unreliable under small samples or skewed marginal distributions. This is the gap that the present project seeks to address.

100 samples of unrelated binary variables

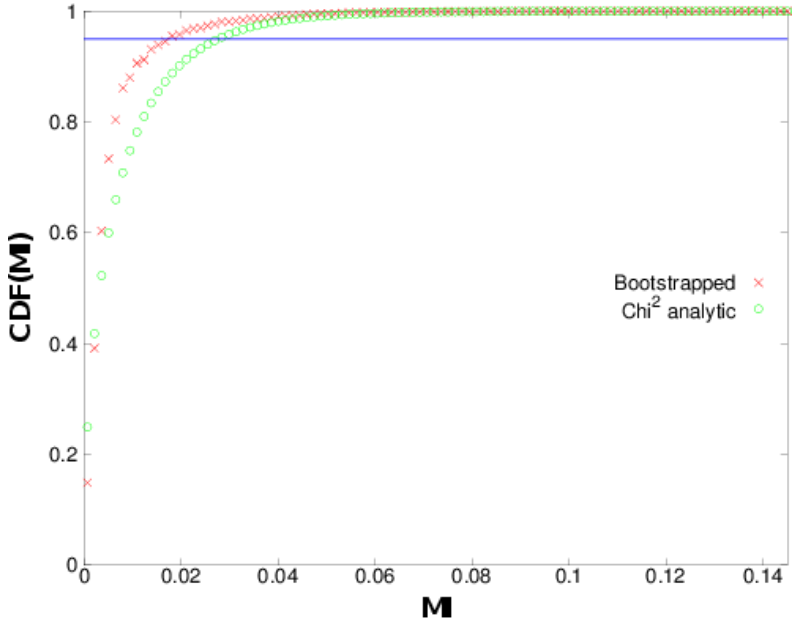
Balanced Case

Each binary variable is distributed with p(0) = 0.5, p(1) = 0.5



Skewed Case

Each binary variable is distributed with p(0) = 0.05, p(1) = 0.95



Research Problem

To derive a corrected finite-sample null distribution for discrete mutual information, and to determine whether it improves on the standard χ^2 null distribution.

Literature Review: The Issues

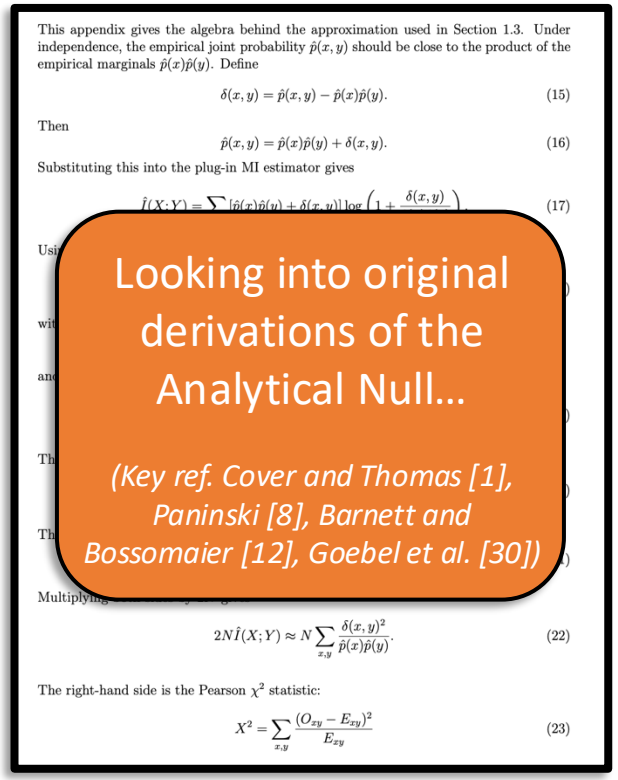
We reviewed the existing literature on the derivation of the analytical null distribution for mutual information and identified two main issues with the current approaches.

Original Form

Analytical Null

$$2N\hat{I}(X;Y) \xrightarrow{d} \chi^2_{(|X|-1)(|Y|-1)}$$

Critical Literature Review



This appendix gives the algebra behind the approximation used in Section 1.3. Under independence, the empirical joint probability $\hat{p}(x, y)$ should be close to the product of the empirical marginals $\hat{p}(x)\hat{p}(y)$. Define

$$\delta(x, y) = \hat{p}(x, y) - \hat{p}(x)\hat{p}(y). \quad (15)$$

Then

$$\hat{p}(x, y) = \hat{p}(x)\hat{p}(y) + \delta(x, y). \quad (16)$$

Substituting this into the plug-in MI estimator gives

$$\hat{I}(X;Y) = \sum [\hat{p}(x)\hat{p}(y) + \delta(x, y)] \log\left(1 + \frac{\delta(x, y)}{\hat{p}(x)\hat{p}(y)}\right). \quad (17)$$

Using $\log(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots$ and keeping the leading term, we obtain

$$2N\hat{I}(X;Y) \approx N \sum_{x,y} \frac{\delta(x, y)^2}{\hat{p}(x)\hat{p}(y)}. \quad (22)$$

The right-hand side is the Pearson χ^2 statistic:

$$X^2 = \sum_{x,y} \frac{(O_{xy} - E_{xy})^2}{E_{xy}} \quad (23)$$

Looking into original derivations of the Analytical Null…

(Key ref. Cover and Thomas [1], Paninski [8], Barnett and Bossomaier [12], Goebel et al. [30])

Main Issues Discovered

Binomial Distr. Approximated to Normal Distr. → **This approximation only works well in large sample regime**

Second-order Taylor expansion around independence → **Works poorly when variable X and Y are heavily skewed and under sampled**

Literature Review: Potential Solutions

We reviewed classical statistics literature and identified three methods that offer potential pathways for deriving a more accurate null distribution.

	A *Exact Combinatorial Enumeration*	B *Effective Degrees of Freedom*	C *Moment-Matching and Parametric Approximation*
Idea	Enumerate every contingency table consistent with the observed marginals	Keep χ^2 shape; replace nominal df with effective df from observed marginals	Match analytical moments of MI to a parametric family (gamma)
Key work	Fisher's exact test [22]; Kirkley [25]	Cliff et al. [26]; Satterthwaite [27]	Goebel et al. [30]
Strength	Exact null, no asymptotic approximation	Reuses familiar χ^2 machinery	Closed-form, fast, no resampling
Gap	Combinatorial explosion beyond small 2×2 tables	Unclear if changing df is sufficient in a skewed, finite-sample regime	Marginal shape drops out due to truncating at 2nd order Taylor Approximation

Approach & Contributions

Building on the literature, we propose an approach to derive a corrected null distribution and outline three contributions that make our approach both practical and accessible

Approach

Primary route

Derive an *exact* closed-form finite-sample null distribution

Fallback Route

Derive an approximate *correction* to the current analytical null distribution

Contributions

1. **A "better" analytical null distribution**
2. **Regime characterisation**
3. **JIDT toolkit integration**

Evaluation Plan

Our corrected null will be evaluated across 45 input configurations and benchmarked against both the chi-square analytical null and the resampled null.

Evaluation Method

Compare Three Methods

Corrected Null
(from this project)

χ^2 Analytical Null
(Baseline to Beat)

Resampled Null
(Ground Truth)

Evaluation Metrics

CDF overlay

do the null distributions match in shape?

Q–Q plot

do quantiles agree, especially in the tail used for thresholds?

False Positive Rate

does each threshold give FPR = nominal α for p-value tests

Test Grid · 45 Configurations

1. **Alphabet size**
   2, 3, 6 outcomes

2. **Sample size**
   N = 20, 50, 100, 200, 500

3. **Marginal skewness**
   balanced, mild (70/30), strong (90/10)

Research Plan

The project will be delivered across five milestones, spanning from Semester 1 Week 10 to Semester 2 Week 13, covering derivation, implementation, evaluation and final write-up.

Sem 1, Week 10



Survey and test the three routes; identify most promising approach

Sem 2, Week 4



Derive and implement the chosen correction in Python

Sem 2, Week 6



Grid-based testing, characterisation of regimes, JIDT integration

Sem 2, Week 8



Thesis Writing and finalisation

Sem 2, Week 13



Finish Thesis!

Appendix: References

Key references from Assignment 3 underpinning the literature review, methodology and the three potential approaches presented in this presentation.

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